

RiskRadar — ML / Data Engineering Progress Update

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Author: Ire (ML / Data Engineer) **Covers:** Dataset selection, repository setup, and geographic + rule-based feature engineering **Status:** First addition — baseline model training will follow as a second update

1. Overview

This update covers everything done on the ML/Data side before model training begins: choosing and validating the dataset, setting up the shared GitHub repository, and building out the fraud-detection rule logic that the model will be trained alongside.

The approach throughout has been: **don't assume a rule works — test it against the real data first, and only fill gaps with clearly-labeled synthetic data where the real data genuinely doesn't cover something.**

2. Dataset

Source: [Nigerian Financial Transactions and Fraud Detection Dataset](#) (HuggingFace)

- 5,000,000 transactions, 45 original columns
- Includes `is_fraud` (boolean) and `fraud_type` (labeled: Account Takeover, Identity Fraud, Impossible Travel Fraud, or none)
- Already includes many pre-engineered fraud-relevant fields: spending deviation score, velocity score, geo-anomaly score, device/IP sharing flags, and more — more feature-rich than a typical raw transaction log

Real fraud rate: 3.59% (179,553 fraudulent out of 5,000,000 rows)

3. Repository Setup

- Private GitHub repo created: `riskradar` (team collaborators invited)
- Local project (`riskradar-ml`) structured as:

```
notebooks/  - EDA, feature engineering, model training notebooks
data/       - raw/ and processed/ (excluded from git - too large to commit)
models/     - trained model files (excluded from git)
simulator/  - synthetic transaction generator, incl. rare-case demo scenarios
docs/       - this document and future updates
```

- `.gitignore` configured to exclude datasets, model files, and the virtual environment, so the repo stays lightweight for all collaborators
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4. The Seven Fraud Rules — What We Found

The team defined 7 candidate fraud-detection rules at the start. Each one was tested against the real dataset before being accepted, adjusted, or dropped — rather than assuming they'd all work as originally written.

#	Rule	Outcome
1	Geographic Risk Flagging	✅ Kept — required real-world research + synthetic augmentation (see §5)

2	Behavioral Spending Deviation	✓ Kept — already present in dataset (spending_deviation_score, user_avg_txn_amt, user_std_txn_amt)
3	Temporal Anomalies (Odd Hours)	✓ Kept — already present in dataset (txn_hour, is_night_txn)
4	Transaction Velocity	✓ Kept — already present in dataset (velocity_score, txn_count_last_1h/24h)
5	Device & Account Takeover (ATO)	✓ Kept, simplified — see §6
6	Impossible Travel	✓ Kept, but rare in real data — see §7
7	Anomalous Airtime Top-ups	✗ Dropped — see §8

5. Rule 1 — Geographic Risk: Research + Augmentation

Problem found: the real dataset only contains transactions from 10 Nigerian cities (mapping to 10 states). None of our team's originally identified high-risk states — Zamfara, Borno, Nasarawa, Plateau — appeared anywhere in the real data.

Research done: looked up real Nigerian fraud statistics to ground our approach in fact rather than guesswork. Confirmed via NIBSS-linked reporting that:

- Lagos accounts for ~63% of national fraud volume (matches real data)
- FCT (Abuja), Ogun, Rivers, and Delta follow, in that order
- Within Lagos specifically, Ikeja, Lagos Island, and Lekki are the top hotspots — Lekki in particular is repeatedly named as a hub for internet/financial fraud

What we built:

- Mapped all 10 real cities to their correct states
- Broke Lagos down into 7 real LGAs (Ikeja, Lagos Island, Lekki, Oshodi, Ikorodu, Apapa, Victoria Island), weighted to match real-world hotspot concentration, with Lekki specifically flagged as high-risk (`is_high_risk_lga`)
- Generated synthetic transactions for the 27 missing states (all 36 states + FCT are now represented), with **volume weighted to match real fraud statistics** (Ogun, Rivers, Delta got proportionally more synthetic rows than other missing states, matching their real reported share)
- Every synthetic row is tagged `source = 'synthetic'` so it can always be distinguished from real data in evaluation
- Synthetic rows were built by **resampling real transaction patterns** (amount, hour, velocity, device behavior) and only relabeling the state — not inventing arbitrary "obviously fraudulent" values. This avoids the model trivially learning "state X = fraud" instead of real fraud behavior.

Result: high-risk states show a real, non-trivial fraud rate lift (7.17% vs 3.59% baseline) — a usable signal without being an artificial giveaway.

Documented limitation: rows for states with no real data are synthetic and should be reported as such; model performance should be evaluated separately on real vs. synthetic data to avoid overstating accuracy.

6. Rule 5 — Account Takeover (ATO)

Original rule required: new device + within 5 minutes of a login/password reset + high-value transfer out.

Problem found: the dataset has no login-event data at all — only transaction timestamps. Testing the full rule (using `time_since_last_transaction` as a stand-in for "time since login") caught only **0.18%** of real labeled ATO cases — essentially useless.

Diagnosis: tested each candidate field individually against the real labeled ATO cases. Found that `new_device_transaction` alone perfectly identifies **100%** of real ATO cases — and every single fraud case tied to this label has this field set to `True` (fraud rate: 0% when `False`, 4.39% when `True`).

Decision: simplified Rule 5 to just `new_device_transaction == True`. Adding transaction-type or amount conditions on top only hurt recall (down to 1.95%), confirming those extra conditions weren't supported by the actual data.

Documented limitation: this dataset ties ATO almost entirely to device novelty; this is a strong signal *in this dataset* but may not generalize to real-world behavior, where legitimate customers also use new devices often.

7. Rule 6 — Impossible Travel

Found: the dataset already includes a pre-built `geospatial_velocity_anomaly` flag. Validated it against the real 'Impossible Travel Fraud' label:

- **100% recall** — catches every real labeled case
- Low precision (3%) — but only because there are just **12 real labeled cases** in the entire 5,086,000-row dataset. This is expected at that sample size, not a flaw in the rule.

Decision: kept as-is. With only 12 real examples, no further tuning would be meaningful — the rule is a reliable supplementary signal for the model, not a standalone high-confidence detector.

Demo consideration: because only 12 real cases exist, this is highly unlikely to appear live during a demo. Built a hand-crafted scenario generator (`generate_impossible_travel_scenario()` in `simulator/generate.py`) to reliably trigger this rule on demand for presentations — clearly separate from the training data.

8. Rule 7 — Anomalous Airtime Top-ups (Dropped)

Problem found: unlike the other rules, this one had **no real support at all** in the dataset:

- Fraud rate for airtime/data transactions (3.54%) was not elevated vs. the overall rate (3.60%) — if anything, slightly lower
- Amount distributions for fraudulent vs. non-fraudulent airtime transactions were statistically indistinguishable
- The amounts themselves were unrealistic for airtime purchases — the dataset's "airtime" transactions averaged ₦743,673, when real MTN/Airtel top-ups are typically ₦100–₦20,000. The category label exists, but the dataset doesn't actually model real airtime economics distinctly from other transaction types.

Decision: Rule 7 is dropped from the trained model. There is no real data to train or validate it against, and building it would mean the model learns nothing genuine — only what we personally invented. Per our team's data-honesty standard (documented in this project from the start), we do not present a rule as data-driven when it isn't.

A lightweight version remains available only as a **scripted simulator demo**

(`generate_airtime_fraud_scenario()`) for illustrative/presentation purposes, but it will **not** be part of the

trained model or evaluated as a real detection capability.

9. Summary of Final Rule Set Going Into Model Training

Rule	Data-driven?	Confidence
1. Geographic Risk	Real + documented synthetic augmentation	High
2. Spending Deviation	Fully real	High
3. Odd Hours	Fully real	High
4. Velocity	Fully real	High
5. Account Takeover (simplified)	Fully real	High
6. Impossible Travel	Fully real (very small sample)	Medium — small sample size
7. Airtime Top-ups	Dropped — no real support	N/A

10. What's Next (Second Addition)

The next update will cover:

- Saving the final cleaned/augmented dataset (`data/processed/nigerian_transactions_clean.csv`)
- Baseline model training (Random Forest, `class_weight='balanced'`)
- Model evaluation — **reported separately for real vs. synthetic data**, per the data-honesty approach established in this document
- Feature importance review, to confirm the model isn't over-relying on any single field (e.g. checking it isn't just keying off `new_device_transaction`)